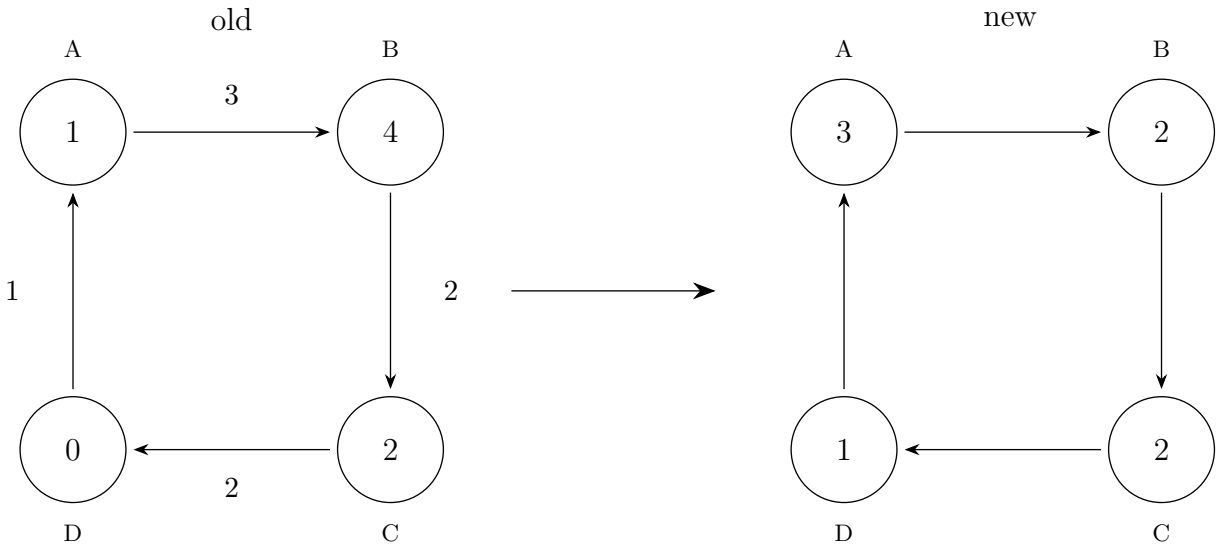
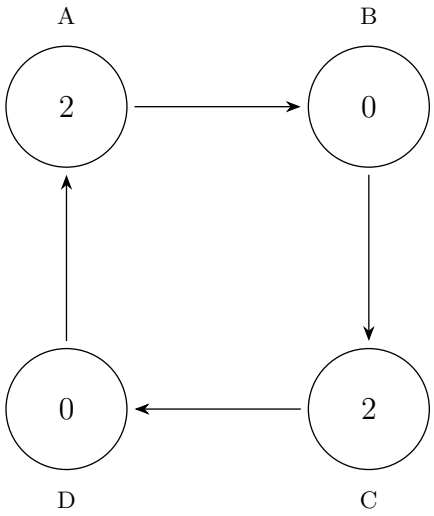
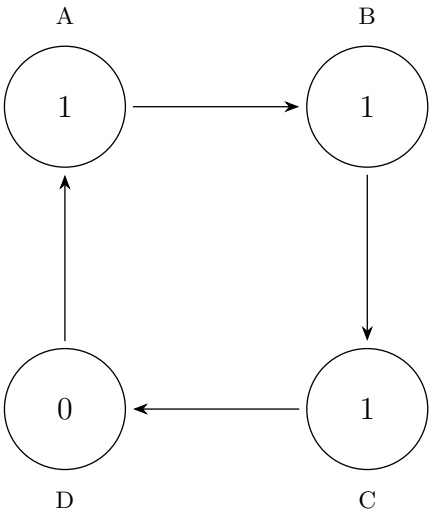
Use whole-number heights, including 0. Follow each arrow: the gap between its two old heights becomes the new height at its tail. Make every new tower before changing any old tower.

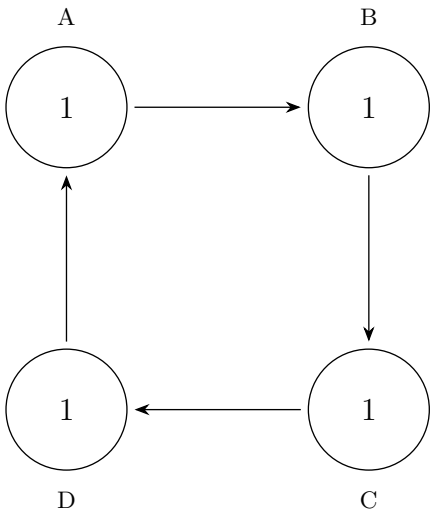


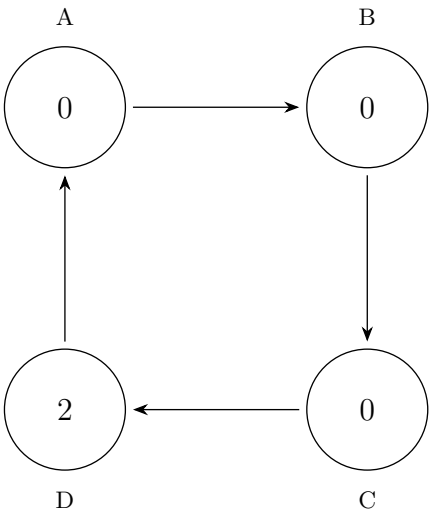
Edge A to B has gap 3, so new A is 3. All four gaps use the old ring.

Problem 3: Could each ring be the result of one move? Find an old ring that makes it, or explain why no old ring can work.

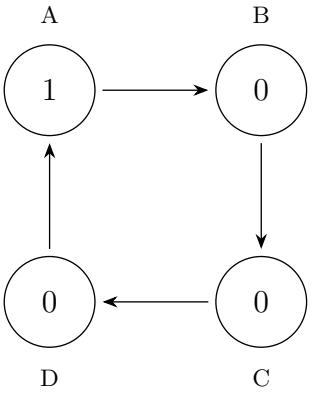


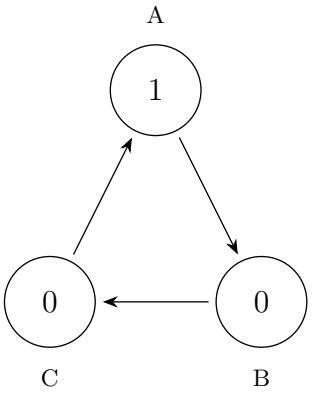






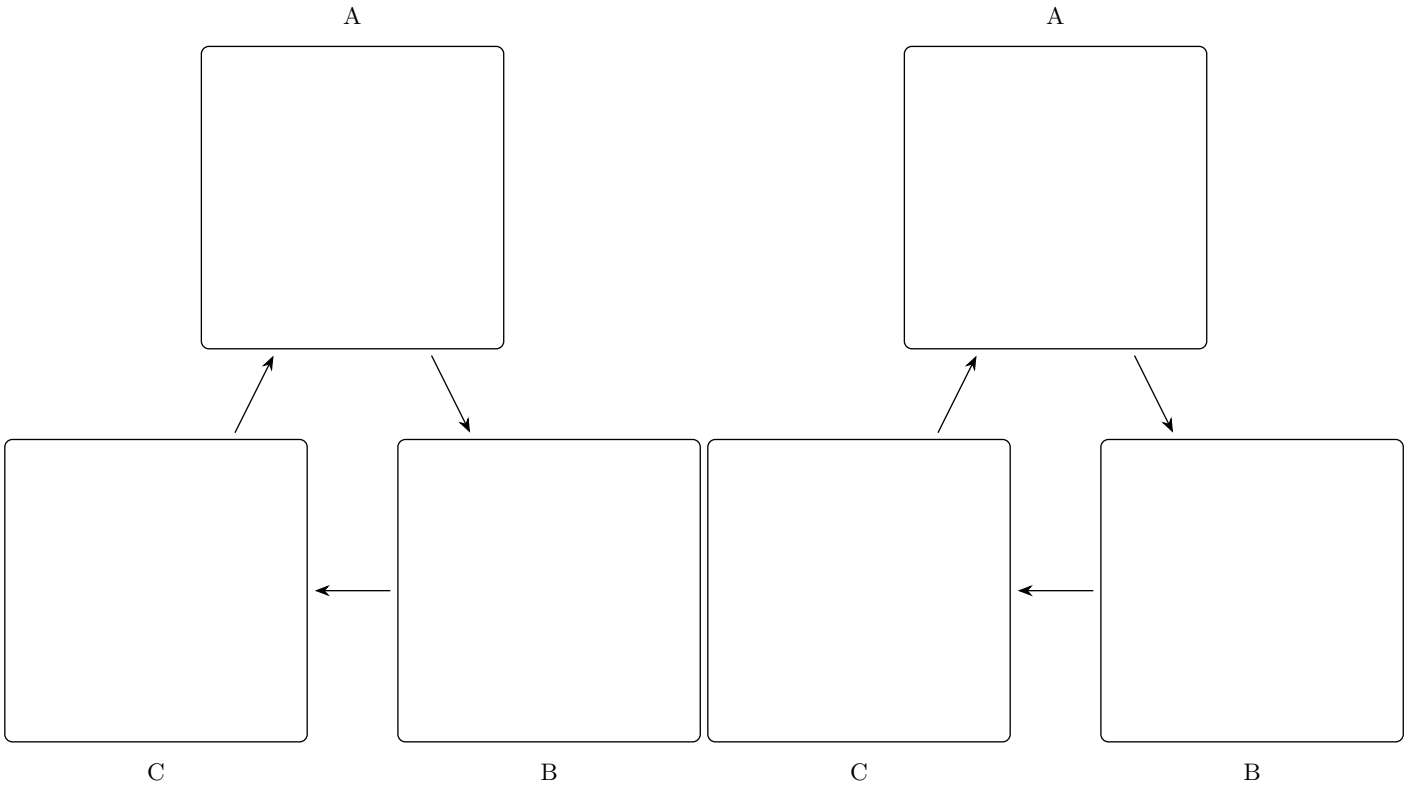
Problem 4: Use the gap rule on both rings. Keep going until the whole ring repeats or becomes all zeros. Does the number of stations change what can happen?





old

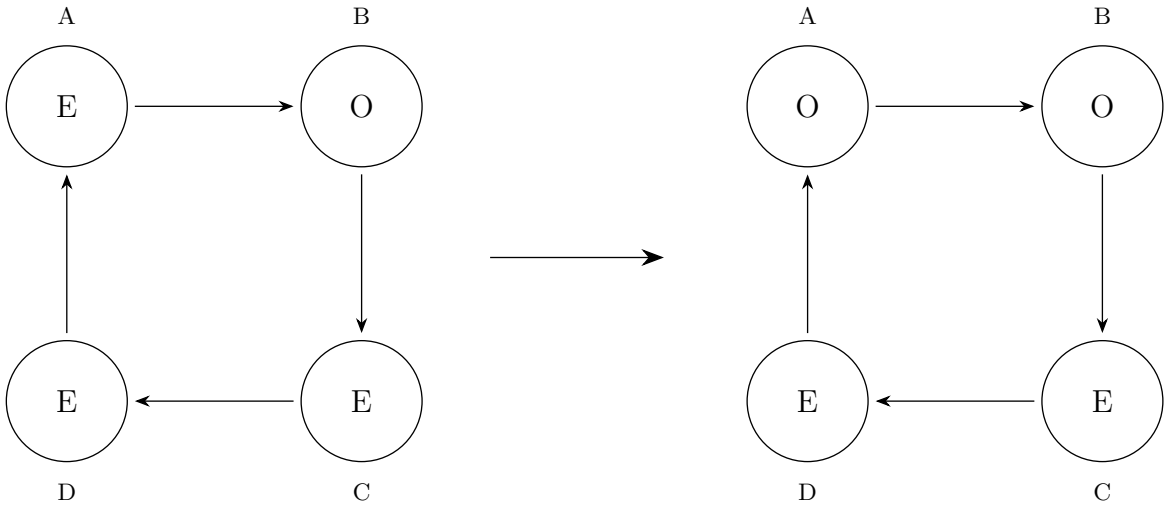
new

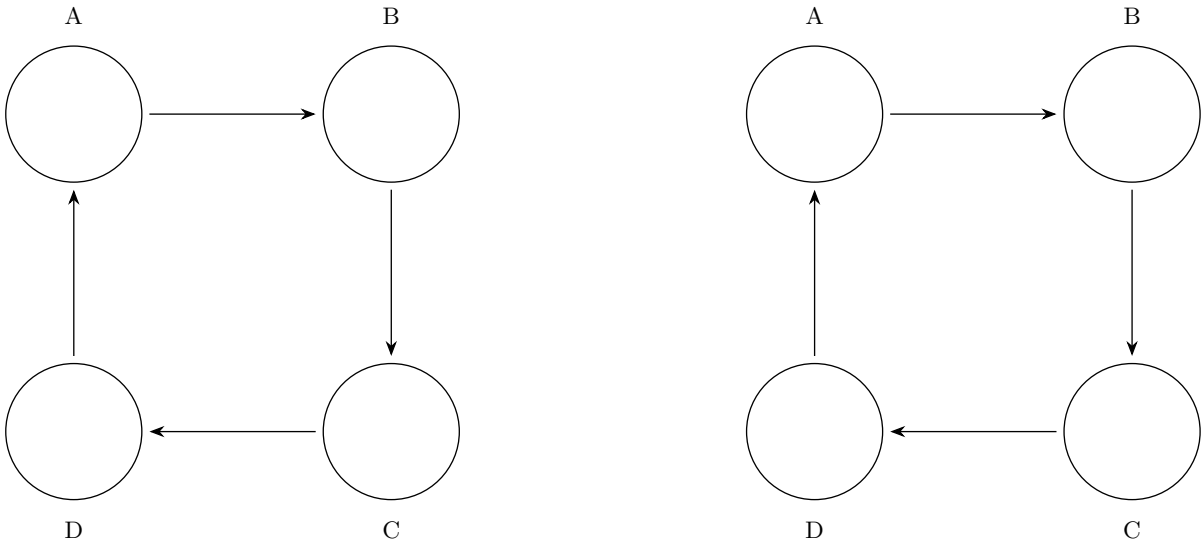


Problem 5: Replace even heights by E and odd heights by O. Use this two-letter gap rule. Which letter rings are possible after four moves? Explain why you have checked every kind of start.

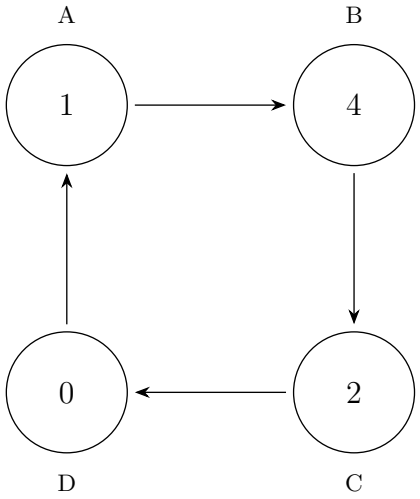
E and E give E. O and O give E.

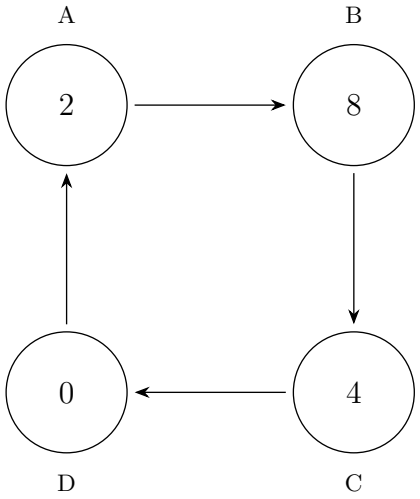
E and O give O. O and E give O.





Problem 6: Compare these two starts through several moves. What happens if all starting heights are doubled? Does your explanation also work for tripling?





Problem 7: Could any new height exceed the largest starting height? Explain why your answer holds however many moves are made.

Problem 8: Do four nonnegative whole-number heights always reach all zeros? Explain why every such start finishes, or find a start that never reaches zero.